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Contour extraction device, contour extraction method, and contour extraction program

Abstract

A contour extraction device propagates a contour of a region of interest of a first tomographic image in a direction from the first tomographic image to a second tomographic image to extract first contours of the regions of interest of tomographic images between the first tomographic image and the second tomographic image, and propagates a contour of the region of interest of the second tomographic image in a direction from the second tomographic image to the first tomographic image to extract second contours of the regions of interest of tomographic images between the first tomographic image and the second tomographic image, and outputs a correction instruction for a contour in a third tomographic image between the first tomographic image and the second tomographic image in a case where a difference between the first contours and the second contours in the third tomographic image is a threshold value or more.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a Continuation of PCT International Application No. PCT/JP2020/040129 filed on Oct. 26, 2020, which claims priority under 35 U.S.C § 119(a) to Japanese Patent Application No. 2019-226791 filed on Dec. 16, 2019. Each of the above application(s) is hereby expressly incorporated by reference, in its entirety, into the present application.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The present disclosure relates to a non-transitory computer readable recording medium storing a

contour extraction device, a contour extraction method, and a contour extraction program.

2. Description of the Related Art

(2) An ultrasonic diagnostic apparatus that extracts a contour of an object in a tomographic image by performing predetermined certain processing on the tomographic image is disclosed (see JP2002-224116A). This ultrasonic diagnostic apparatus performs interpolation processing using two contours extracted from tomographic images having two different time points in the past to extract a contour temporally positioned between the two contours from the tomographic image.

SUMMARY OF THE INVENTION

(3) In the technique disclosed in JP2002-224116A, the contour may not always be extracted accurately because the ultrasonic diagnostic apparatus performs interpolation processing to extract the contour temporally positioned between the two contours from the tomographic image.

(4) The present disclosure has been made in view of the above circumstances, and provides a contour extraction device, a contour extraction method, and a non-transitory computer readable recording medium storing a contour extraction program capable of accurately extracting a contour of a region of interest from a tomographic image.

(5) There is provided a contour extraction device of the present disclosure comprising: at least one processor, in which the processor propagates a contour of a region of interest of a first tomographic image in a three-dimensional image, which is formed of a plurality of tomographic images, each including the region of interest, in a direction from the first tomographic image to a second tomographic image to extract first contours of the regions of interest of tomographic images between the first tomographic image and the second tomographic image, and propagates a contour of the region of interest of the second tomographic image in a direction from the second tomographic image to the first tomographic image to extract second contours of the regions of interest of tomographic images between the first tomographic image and the second tomographic image, and outputs a correction instruction for a contour in a third tomographic image between the first tomographic image and the second tomographic image in a case where a difference between the first contours and the second contours in the third tomographic image is a threshold value or more.

(6) In the contour extraction device of the present disclosure, the processor may output the extracted contour of each tomographic image in a case where the difference is less than the threshold value.

(7) Further, in the contour extraction device of the present disclosure, the processor, in a case where the difference is the threshold value or more, may extract the first contours and the second contours using a contour corrected in accordance with the correction instruction as the contour of the region of interest of the second tomographic image, in a tomographic image group from the first tomographic image to the third tomographic image, and may extract the first contours and the second contours using the contour corrected in accordance with the correction instruction as the contour of the region of interest of the first tomographic image, in a tomographic image group from the third tomographic image to the second tomographic image.

(8) Further, in the contour extraction device of the present disclosure, the third tomographic image may be a middle tomographic image between the first tomographic image and the second tomographic image in the plurality of tomographic images.

(9) Further, in the contour extraction device of the present disclosure, the processor may extract the first contours and the second contours using an active contour method.

(10) Further, there is provided a contour extraction method of the present disclosure that is executed by a processor provided in a contour extraction device, the method comprising: propagating a contour of a region of interest of a first tomographic image in a three-dimensional image, which is formed of a plurality of tomographic images, each including the region of interest, in a direction from the first tomographic image to a second tomographic image to extract first contours of the regions of interest of tomographic images between the first tomographic image and

the second tomographic image, and propagating a contour of the region of interest of the second tomographic image in a direction from the second tomographic image to the first tomographic image to extract second contours of the regions of interest of tomographic images between the first tomographic image and the second tomographic image; and outputting a correction instruction for a contour in a third tomographic image between the first tomographic image and the second tomographic image in a case where a difference between the first contours and the second contours in the third tomographic image is a threshold value or more.

(11) Further, there is provided a non-transitory computer readable recording medium storing a contour extraction program of the present disclosure for causing a processor provided in a contour extraction device to execute a process comprising: propagating a contour of a region of interest of a first tomographic image in a three-dimensional image, which is formed of a plurality of tomographic images, each including the region of interest, in a direction from the first tomographic image to a second tomographic image to extract first contours of the regions of interest of tomographic images between the first tomographic image and the second tomographic image, and propagating a contour of the region of interest of the second tomographic image in a direction from the second tomographic image to the first tomographic image to extract second contours of the regions of interest of tomographic images between the first tomographic image and the second tomographic image; and outputting a correction instruction for a contour in a third tomographic image between the first tomographic image and the second tomographic image in a case where a difference between the first contours and the second contours in the third tomographic image is a threshold value or more.

(12) According to the present disclosure, it is possible to accurately extract a contour of a region of interest from a tomographic image.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a block diagram showing an example of a configuration of a diagnosis support system.

(2) FIG. 2 is a block diagram showing an example of a hardware configuration of a contour extraction device.

(3) FIG. 3 is a block diagram showing an example of a functional configuration of the contour extraction device.

(4) FIG. 4 is a view illustrating contours of regions of interest of first tomographic image and second tomographic image.

(5) FIG. 5 is a view illustrating contour extraction processing.

(6) FIG. 6 is a view illustrating the contour extraction processing.

(7) FIG. 7 is a flowchart showing an example of the contour extraction processing.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

(8) Hereinafter, an exemplary embodiment of the technique of the present disclosure will be described in detail with reference to the drawings.

(9) First, the configuration of a diagnosis support system **10** according to the present embodiment will be described with reference to FIG. 1. As shown in FIG. 1, the diagnosis support system **10** includes a contour extraction device **12**, an imaging device **14**, and an image storage device **16**. The contour extraction device **12**, the imaging device **14**, and the image storage device **16** are each connected to a network N and can communicate with each other through the network N.

(10) The imaging device **14** is a device that generates a three-dimensional medical image representing a part to be diagnosed of an object to be examined by imaging the part. The three-dimensional medical image captured by the imaging device **14** is formed of a plurality of tomographic images. In the present embodiment, an example in which a computed tomography

(CT) device is applied as the imaging device **14** will be described, but the present disclosure is not limited thereto. For example, a device that generates three-dimensional medical images other than a CT device, such as a magnetic resonance imaging (MRI) device or a positron emission tomography (PET) device, may be applied as the imaging device **14**. Further, in the present embodiment, an example in which a tomographic image of an axial cross-section is applied as a tomographic image constituting the three-dimensional medical image will be described, but the present disclosure is not limited thereto. A tomographic image of a cross-section other than the axial cross-section, such as a sagittal cross-section and a coronal cross-section, may be applied as the tomographic image constituting the three-dimensional medical image.

(11) The image storage device **16** is a computer that stores and manages medical images, and comprises, for example, a storage device that stores medical images. The image storage device **16** transmits/receives the medical images generated by the imaging device **14** between the contour extraction device **12** and the imaging device **14** through the network N. The storage format of the medical image and the communication between devices through the network N are based on a predetermined protocol, such as digital imaging and communications in medicine (DICOM) or the like.

(12) Next, the hardware configuration of the contour extraction device **12** according to the present embodiment will be described with reference to FIG. 2. Examples of the contour extraction device **12** include a personal computer or a server computer or the like. The contour extraction device **12** may be a cloud server.

(13) As shown in FIG. 2, the contour extraction device **12** includes a central processing unit (CPU) **20**, a memory **21** as a temporary storage area, and a nonvolatile storage unit **22**. The contour extraction device **12** includes a display **23**, such as a liquid crystal display, an input unit **24**, such as a keyboard and a mouse, and a network interface (I/F) **25** connected to the network N. The CPU **20**, the memory **21**, the storage unit **22**, the display **23**, the input unit **24**, and the network I/F **25** are connected to a bus **26**.

(14) The storage unit **22** is realized by a hard disk drive (HDD), a solid state drive (SSD), a flash memory, or the like. The storage unit **22** as a storage medium stores a contour extraction program **30**. The CPU **20** reads the contour extraction program **30** from the storage unit **22**, loads the contour extraction program **30** into the memory **21**, and executes the loaded contour extraction program **30**.

(15) Next, the functional configuration of the contour extraction device **12** according to the present embodiment will be described with reference to FIG. 3. As shown in FIG. 3, the contour extraction device **12** includes a reception unit **40**, an extraction unit **42**, and an output unit **44**. The CPU **20** executes the contour extraction program **30** to function as the reception unit **40**, the extraction unit **42**, and the output unit **44**.

(16) As shown in FIG. 4, a user such as a doctor selects, for example, two tomographic images that are tomographic images positionally located at both ends among a tomographic image group, which is an extraction target of a region of interest, from among the plurality of tomographic images, each including the region of interest, the plurality of tomographic images constituting the three-dimensional medical image captured by the imaging device **14**. The two selected tomographic images are different tomographic images from each other, and there are a plurality of tomographic images between the two selected tomographic images. The region of interest (ROI) as used herein means a region of interest to the user, for example, a region of a lesion, such as a tumor, and a region of an organ.

(17) Hereinafter, among the two tomographic images selected by the user, one tomographic image is referred to as a “first tomographic image”, and the other tomographic image is referred to as a “second tomographic image”. Since the accuracy of contour extraction processing of the region of interest, which will be described later, may decrease in a case where the regions of interest included in the first and the second tomographic images are too small, the user selects tomographic images,

each of which includes a region of interest having a certain size or larger in the present embodiment.

(18) Next, the user designates a contour of the region of interest of the first tomographic image and a contour of the region of interest of the second tomographic image through the input unit **24**. The reception unit **40** receives the contour of the region of interest of the first tomographic image and the contour of the region of interest of the second tomographic image designated by the user. Further, the reception unit **40** receives a contour designated by the user in accordance with a correction instruction output by the output unit **44**, which will be described later.

(19) The extraction unit **42** propagates the contour of the region of interest of the first tomographic image, which is received by the reception unit **40**, in a direction from the first tomographic image to the second tomographic image to extract first contours **O1** of the regions of interest of tomographic images between the first and the second tomographic images. In the present embodiment, as shown in FIG. 5, the extraction unit **42** propagates the contour of the region of interest of the first tomographic image from the first tomographic image to a third tomographic image that is a middle tomographic image between the first and the second tomographic images. With this, the extraction unit **42** extracts the first contours **O1** of regions of interest of the third tomographic image and of the tomographic image between the first and the third tomographic images. In FIG. 5, the first contour **O1** is shown by an alternating long-dash and short-dash line. For example, the extraction unit **42** may extract the first contours **O1** from all the tomographic images between the first and the second tomographic images.

(20) Further, the extraction unit **42** propagates the contour of the region of interest of the second tomographic image, which is received by the reception unit **40**, in a direction from the second tomographic image to the first tomographic image to extract second contours **O2** of the regions of interest of tomographic images between the first and the second tomographic images. In the present embodiment, as shown in FIG. 5, the extraction unit **42** propagates the contour of the region of interest of the second tomographic image from the second to the third tomographic images. With this, the extraction unit **42** extracts the second contours **O2** of the regions of interest of the third tomographic image and of the tomographic image between the second and the third tomographic images. In FIG. 5, the second contour **O2** is shown by an alternating long-dash and two-short-dash line. For example, the extraction unit **42** may extract the second contours **O2** from all the tomographic images between the first and the second tomographic images.

(21) The extraction unit **42** according to the present embodiment uses an active contour method, such as a level set method, for the above-described propagation of the region of interest of the first tomographic image and the above-described propagation of the region of interest of the second tomographic image to extract the first contours **O1** and the second contours **O2**.

(22) As shown in FIG. 5, the output unit **44** outputs the contour of each tomographic image extracted by the extraction unit **42** to the storage unit **22**, in a case where the difference between the first contour **O1** and the second contour **O2**, which are extracted by the extraction unit **42**, in the third tomographic image is less than a threshold value **TH**. With this, the contour of each tomographic image is stored in the storage unit **22**. The threshold value **TH**, in this case, is predetermined according to, for example, the required contour extraction accuracy. Further, examples of the difference between the first contour **O1** and the second contour **O2** include any one of a difference in area, a difference in curvature, or a difference in center of gravity between the first contour **O1** and the second contour **O2**, or a combination of two or more of the above-described examples. The output unit **44** may output the contour of each tomographic image extracted by the extraction unit **42** to the display **23** to make the display **23** display the contour.

(23) On the other hand, the output unit **44** outputs the correction instruction for a contour in the third tomographic image to the display **23** in a case where the difference between the first contour **O1** and the second contour **O2** in the third tomographic image is the threshold value **TH** or more. Specifically, in this case, for example, the output unit **44** outputs the third tomographic image and

at least one of the first contour O1 or the second contour O2 to the display 23 to make the display 23 display the third tomographic image and the at least one contour. The user corrects the contour of the region of interest on the third tomographic image displayed on the display 23, through the input unit 24. As described above, the contour of the region of interest of this third tomographic image is received by the reception unit 40.

(24) In this case, as shown in FIG. 6, the tomographic image group from the first to the second tomographic images is divided into two tomographic image groups, that is, a first tomographic image group from the first to the third tomographic images and a second tomographic image group from the third to the second tomographic images. Then, the extraction unit 42 performs processing for extracting the first contours O1 and the second contours O2 in the first and the second tomographic image groups in the same manner as in the tomographic image group from the first to the second tomographic images. This processing is repeatedly performed until the difference between the first contour O1 and the second contour O2 in the third tomographic image becomes less than the threshold value TH.

(25) That is, the extraction unit 42 extracts the first contours O1 and the second contours O2 using the contour corrected in accordance with the correction instruction output by the output unit 44 as the contour of the region of interest of the second tomographic image, in the first tomographic image group. Further, the extraction unit 42 extracts the first contours O1 and the second contours O2 using the contour corrected in accordance with the correction instruction output by the output unit 44 as the contour of the region of interest of the first tomographic image, in the second tomographic image group.

(26) Next, an operation of the contour extraction device 12 according to the present embodiment will be described with reference to FIG. 7. The CPU 20 executes the contour extraction program 30, whereby the contour extraction processing shown in FIG. 7 is executed. The contour extraction processing is executed, for example, in a case where an execution instruction is input by the user through the input unit 24.

(27) In step S10 of FIG. 7, the reception unit 40 receives the contour of the region of interest of the first tomographic image and the contour of the region of interest of the second tomographic image designated by the user. In step S12, as described above, the extraction unit 42 propagates the contour of the region of interest of the first tomographic image in a direction from the first tomographic image to the second tomographic image to extract the first contours O1 of the regions of interest of the tomographic images between the first and the second tomographic images.

(28) In step S14, as described above, the extraction unit 42 propagates the contour of the region of interest of the second tomographic image in a direction from the second tomographic image to the first tomographic image to extract the second contours O2 of the regions of interest of the tomographic images between the first and the second tomographic images. In step S16, as described above, the output unit 44 determines whether or not the difference between the first contour O1 extracted in step S12 and the second contour O2 extracted in step S14 in the third tomographic image is less than the threshold value TH. In a case where a negative determination is made, the process proceeds to step S18.

(29) In step S18, as described above, the output unit 44 outputs the correction instruction for the contour in the third tomographic image to the display 23. In step S20, the reception unit 40 receives the contour corrected by the user in accordance with the correction instruction output in step S18. When the processing in step S20 ends, the process returns to step S12. In this case, the processing after step S12 is executed using the third tomographic image as the second tomographic image, for the first tomographic image group from the first to the third tomographic images. Further, in this case, the processing after step S12 is executed using the third tomographic image as the first tomographic image, for the second tomographic image group from the third to the second tomographic images.

(30) On the other hand, in a case where an affirmative determination is made in step S16, the

process proceeds to step S22. In step S22, the output unit **44** outputs the contour of each tomographic image to the storage unit **22** as described above. When the processing of step S22 ends, the contour extraction processing ends.

(31) As described above, according to the present embodiment, the contour of the region of interest of the first tomographic image is propagated in the direction from the first tomographic image to the second tomographic image, whereby the first contours of the regions of interest of the tomographic images between the first and the second tomographic images are extracted. Further, the contour of the region of interest of the second tomographic image is propagated in the direction from the second tomographic image to the first tomographic image, whereby the second contours of the regions of interest of the tomographic images between the first and the second tomographic images are extracted. Then, the correction instruction for the contour in the third tomographic image between the first and the second tomographic images is output in a case where the difference between the first and the second contours in the third tomographic image is the threshold value or more. Accordingly, the contour of the region of interest can be accurately extracted from the tomographic image.

(32) In the above-described embodiment, the middle tomographic image between the first and the second tomographic images is applied as the third tomographic image, but the present disclosure is not limited thereto. For example, a tomographic image other than the middle tomographic image between the first and the second tomographic images may be applied as the third tomographic image. In this case, for example, a tomographic image having the largest area of the contour extracted by the extraction unit **42** can be applied as the third tomographic image.

(33) Further, in the above-described embodiment, for example, the following various processors can be used as the hardware structures of processing units that execute various kinds of processing, such as the reception unit **40**, the extraction unit **42**, and the output unit **44**. The various processors include, for example, a programmable logic device (PLD), such as a field programmable gate array (FPGA), which is a processor having a changeable circuit configuration after manufacture, and a dedicated electrical circuit, such as an application specific integrated circuit (ASIC), which is a processor having a dedicated circuit configuration designed to perform specific processing, in addition to the CPU, which is a general-purpose processor that executes software (programs) to function as various processing units, as described above.

(34) One processing unit may be constituted of one of the various processors or may be constituted of a combination of two or more processors of the same type or different types (for example, a combination of a plurality of FPGAs and a combination of a CPU and an FPGA). Alternatively, the plurality of processing units may be constituted of one processor.

(35) A first example of the configuration in which the plurality of processing units are constituted of one processor is an aspect in which one or more CPUs and software are combined to constitute one processor and the processor functions as a plurality of processing units. A representative example of the aspect is a computer such as a client and a server. A second example of the configuration is an aspect in which a processor that implements all of the functions of a system including the plurality of processing units with one integrated circuit (IC) chip is used. A representative example of the aspect is a system on chip (SoC). As described above, various processing units are constituted of one or more of the various processors as the hardware structures.

(36) Furthermore, as the hardware structures of the various processors, more specifically, an electrical circuit (circuitry) in which circuit elements, such as semiconductor elements, are combined can be used.

(37) In the above-described embodiment, the aspect in which the contour extraction program **30** is stored (installed) in the storage unit **22** in advance has been described, but the present disclosure is not limited thereto. The contour extraction program **30** may be provided in a form in which the contour extraction program **30** is recorded on a recording medium, such as a compact disc read only memory (CD-ROM), a digital versatile disc read only memory (DVD-ROM), and a Universal

Serial Bus (USB) memory. Alternatively, the contour extraction program **30** may also be provided in a form in which the contour extraction program **30** is downloaded from an external device through the network.

EXPLANATION OF REFERENCES

(38) **10**: diagnosis support system **12**: contour extraction device **14**: imaging device **16**: image storage device **20**: CPU **21**: memory **22**: storage unit **23**: display **24**: input unit **25**: network I/F **26**: bus **30**: contour extraction program **40**: reception unit **42**: extraction unit **44**: output unit **N**: network **O1**, **O2**: contour

Claims

1. A contour extraction device comprising: at least one processor, wherein the processor propagates a contour of a region of interest of a first tomographic image in a three-dimensional image, which is formed of a plurality of tomographic images, each including the region of interest, in a first direction from the first tomographic image to a second tomographic image to extract first contours of the regions of interest of tomographic images between the first tomographic image and the second tomographic image, and propagates a contour of the region of interest of the second tomographic image in a second direction from the second tomographic image to the first tomographic image to extract second contours of the regions of interest of tomographic images between the first tomographic image and the second tomographic image, wherein the first direction is opposite to the second direction, and outputs a correction instruction for a contour in a third tomographic image between the first tomographic image and the second tomographic image in a case where a difference between the first contours and the second contours in the third tomographic image is a threshold value or more, wherein the processor, in a case where the difference is the threshold value or more, extracts the first contours and the second contours using a contour corrected in accordance with the correction instruction as the contour of the region of interest of the second tomographic image, in a tomographic image group from the first tomographic image to the third tomographic image.
2. The contour extraction device according to claim 1, wherein the processor outputs the extracted contour of each tomographic image in a case where the difference is less than the threshold value.
3. The contour extraction device according to claim 1, wherein the third tomographic image is a middle tomographic image between the first tomographic image and the second tomographic image in the plurality of tomographic images.
4. The contour extraction device according to claim 1, wherein the processor extracts the first contours and the second contours using an active contour method.
5. A contour extraction method that is executed by a processor provided in a contour extraction device, the method comprising: propagating a contour of a region of interest of a first tomographic image in a three-dimensional image, which is formed of a plurality of tomographic images, each including the region of interest, in a first direction from the first tomographic image to a second tomographic image to extract first contours of the regions of interest of tomographic images between the first tomographic image and the second tomographic image, and propagating a contour of the region of interest of the second tomographic image in a second direction from the second tomographic image to the first tomographic image to extract second contours of the regions of interest of tomographic images between the first tomographic image and the second tomographic image, wherein the first direction is opposite to the second direction; and outputting a correction instruction for a contour in a third tomographic image between the first tomographic image and the second tomographic image in a case where a difference between the first contours and the second contours in the third tomographic image is a threshold value or more, in a case where the difference is the threshold value or more, extracting the first contours and the second contours using a contour corrected in accordance with the correction instruction as the contour of the region of interest of the

- second tomographic image, in a tomographic image group from the first tomographic image to the third tomographic image.
6. A non-transitory computer readable recording medium storing a contour extraction program for causing a processor provided in a contour extraction device to execute a process comprising: propagating a contour of a region of interest of a first tomographic image in a three-dimensional image, which is formed of a plurality of tomographic images, each including the region of interest, in a first direction from the first tomographic image to a second tomographic image to extract first contours of the regions of interest of tomographic images between the first tomographic image and the second tomographic image, and propagating a contour of the region of interest of the second tomographic image in a second direction from the second tomographic image to the first tomographic image to extract second contours of the regions of interest of tomographic images between the first tomographic image and the second tomographic image, wherein the first direction is opposite to the second direction; and outputting a correction instruction for a contour in a third tomographic image between the first tomographic image and the second tomographic image in a case where a difference between the first contours and the second contours in the third tomographic image is a threshold value or more, in a case where the difference is the threshold value or more, extracting the first contours and the second contours using a contour corrected in accordance with the correction instruction as the contour of the region of interest of the second tomographic image, in a tomographic image group from the first tomographic image to the third tomographic image.
7. The contour extraction device according to claim 1, wherein the processor extracts the first contours and the second contours using the contour corrected in accordance with the correction instruction as the contour of the region of interest of the first tomographic image, in a tomographic image group from the third tomographic image to the second tomographic image.
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